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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM DRC BUS PT H03																																																															
<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td colspan="2">PROJECT TITLE</td><td rowspan="2">ABB</td><td rowspan="2">EPMV</td><td colspan="2" rowspan="4">COVER SHEET MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM DRC BUS PT</td><td>Scale</td><td></td><td>= H03</td></tr><tr><td></td><td></td><td></td><td></td><td>Prepared</td><td>YR</td><td colspan="2" rowspan="2">MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& 308</td></tr><tr><td>00</td><td>Issued for Approval</td><td>17.01.19</td><td>YR</td><td>Checked</td><td>AW</td><td colspan="2" rowspan="2">Based on: BE318042</td><td colspan="2" rowspan="2">PT. ABB Sakti Industri</td><td>Drawing Nbr.</td><td>18.042-MNA-308.12</td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>																				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	COVER SHEET MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM DRC BUS PT		Scale		= H03					Prepared	YR	MV Switchgear for MNA Serang Plant		Customer	PT Multimas Nabati Asahan	& 308	00	Issued for Approval	17.01.19	YR	Checked	AW	Based on: BE318042		PT. ABB Sakti Industri		Drawing Nbr.	18.042-MNA-308.12	Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Customer Doc. Nbr.		Cont. 16
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					Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-32 20kV Tank Farm WICA		Customer	PT Multimas Nabati Asahan	& 308	
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR						
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR						
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.						
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT						
B	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE						B
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT						
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT						
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT						
	-BGK		POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET						
C	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						C
	-BGS1...-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.						
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE						
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE						
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1						
D	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2						D
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3						
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1						
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2						
E	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						E
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER						
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1						
	-BGV2		POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2						
F	-BM		HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3						F
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES						
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS						
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR						
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR						
								-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
								-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
								-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
								-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
								-KFC	CLOSING RELAYS OR CONTACTORS							
								-KFI	INTEGRATED CIRCUITS							
								-KFL	LOCKOUT RELAY							
								-KFM	MICROPROCESSORS							
								-KFN	ANTIPUMPING RELAYS							
								-KFO	OPENING RELAYS OR CONTACTORS							
								-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
								-KFR	RECLOSING RELAYS							
								-KFS	SYNCHRONIZING DEVICES							
								-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
								-KFZ	CONTROLLERS							
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							

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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

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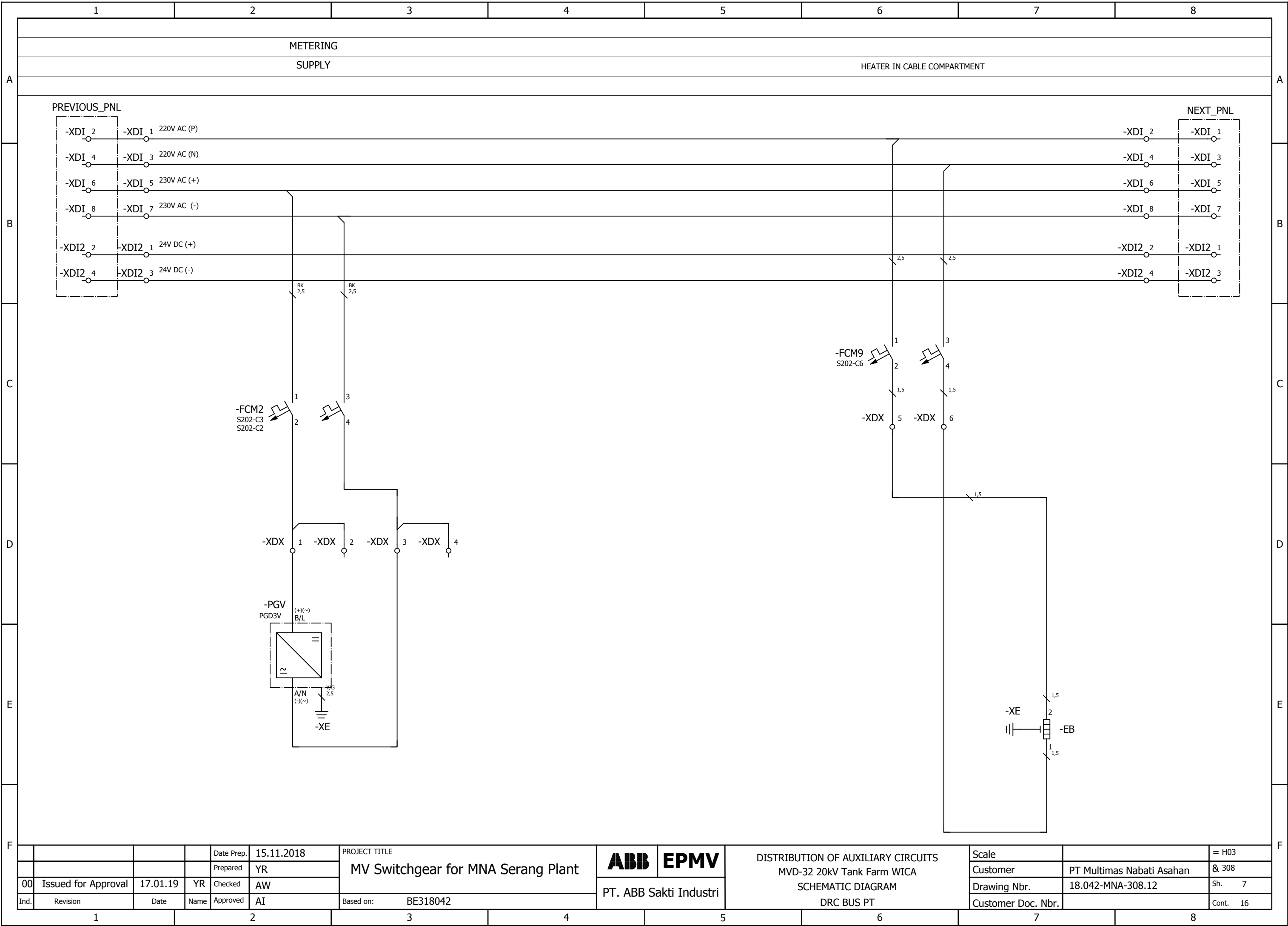
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A	-TFA	ACTIVE POWER TRANSDUCERS	-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A
	-TFC	CURRENT TRANSDUCERS		-XD12	
B	-TFF	FREQUENCY TRANSDUCERS	-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION	B
	-TFJ	ACTIVE ENERGY TRANSDUCERS		-XD14	
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS	
	-TFM	MULTIFUNCTION TRANSDUCERS	-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS	
	-TFP	POWER-FACTOR TRANSDUCERS	-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS	
	-TFQ	REACTIVE POWER TRANSDUCERS	-XD19	TERMINAL BLOCK FOR SPECIAL USE	
	-TFS	SIGNAL CONVERTERS	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD	
	-TFV	VOLTAGE TRANSDUCERS	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS	
	-WA	BUSBARS	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT	
	C	-WBC	POWER CABLES	-XDS	
-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB	
-WGC		CONTROL CABLES	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WH		OPTICAL FIBRES	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS	
-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1	
-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2	
-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3	
-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR	
-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDX	SUPPORT TERMINAL BLOCK	
D	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE	D
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE	
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE	
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE	
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT	
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR			
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			
	-XDC	CUSTOMER TERMINAL BLOCK			
E	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			E
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR			
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS			
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			
F	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			F
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS			
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS			
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)			
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)			

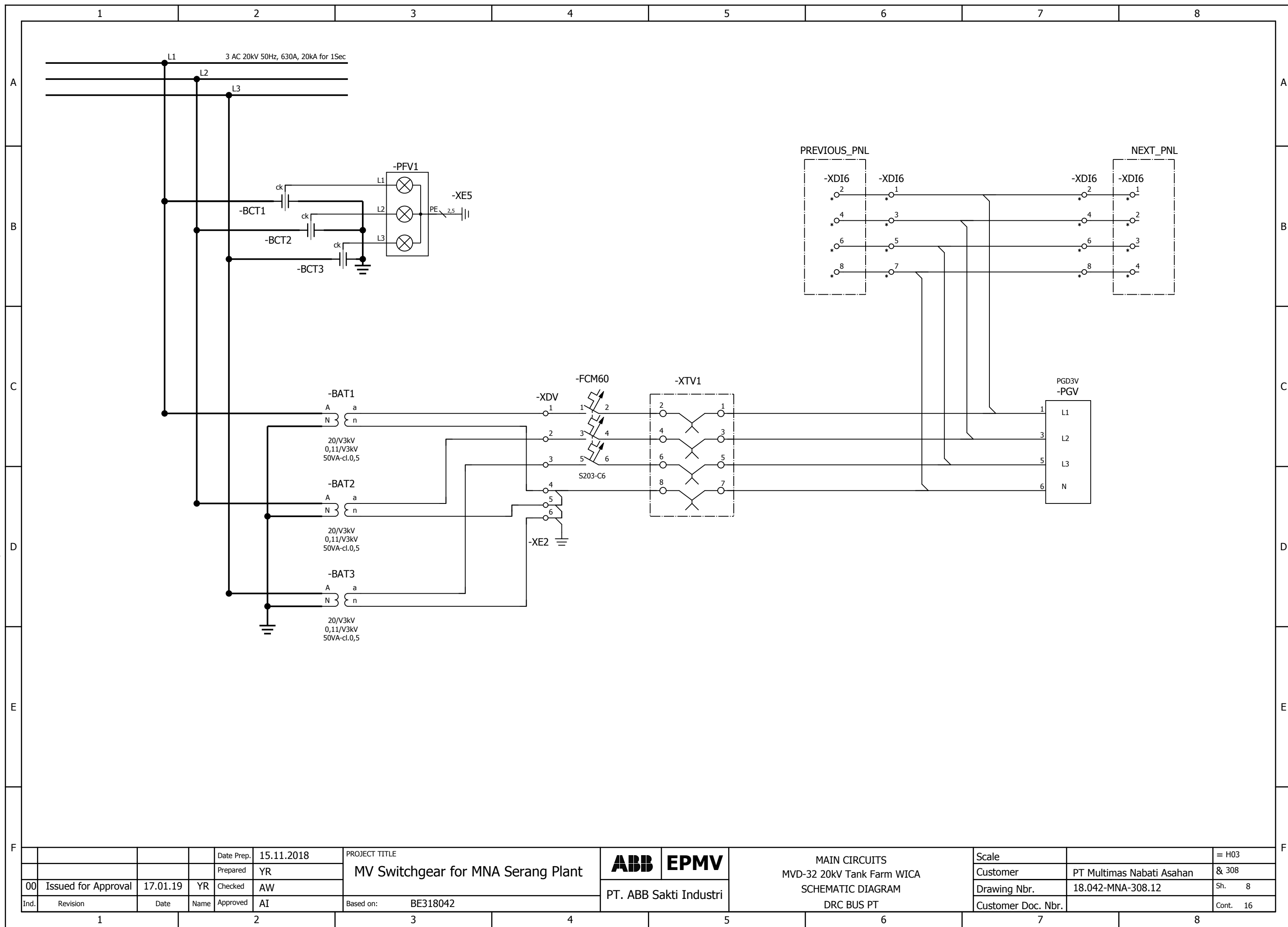
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Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				DRC BUS PT	Customer Doc. Nbr.		Cont. 16

1	2	3	4	5	6	7	8
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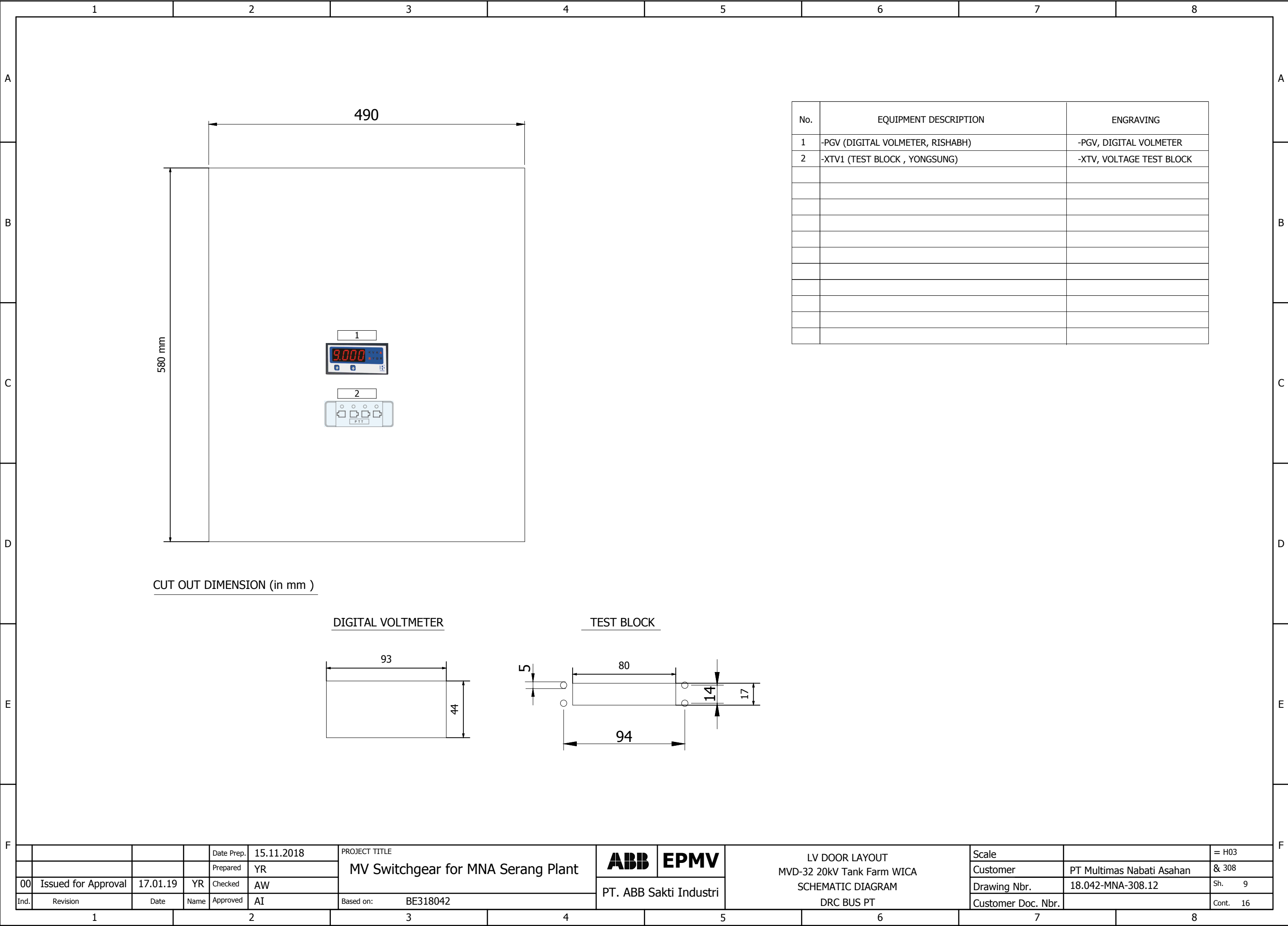
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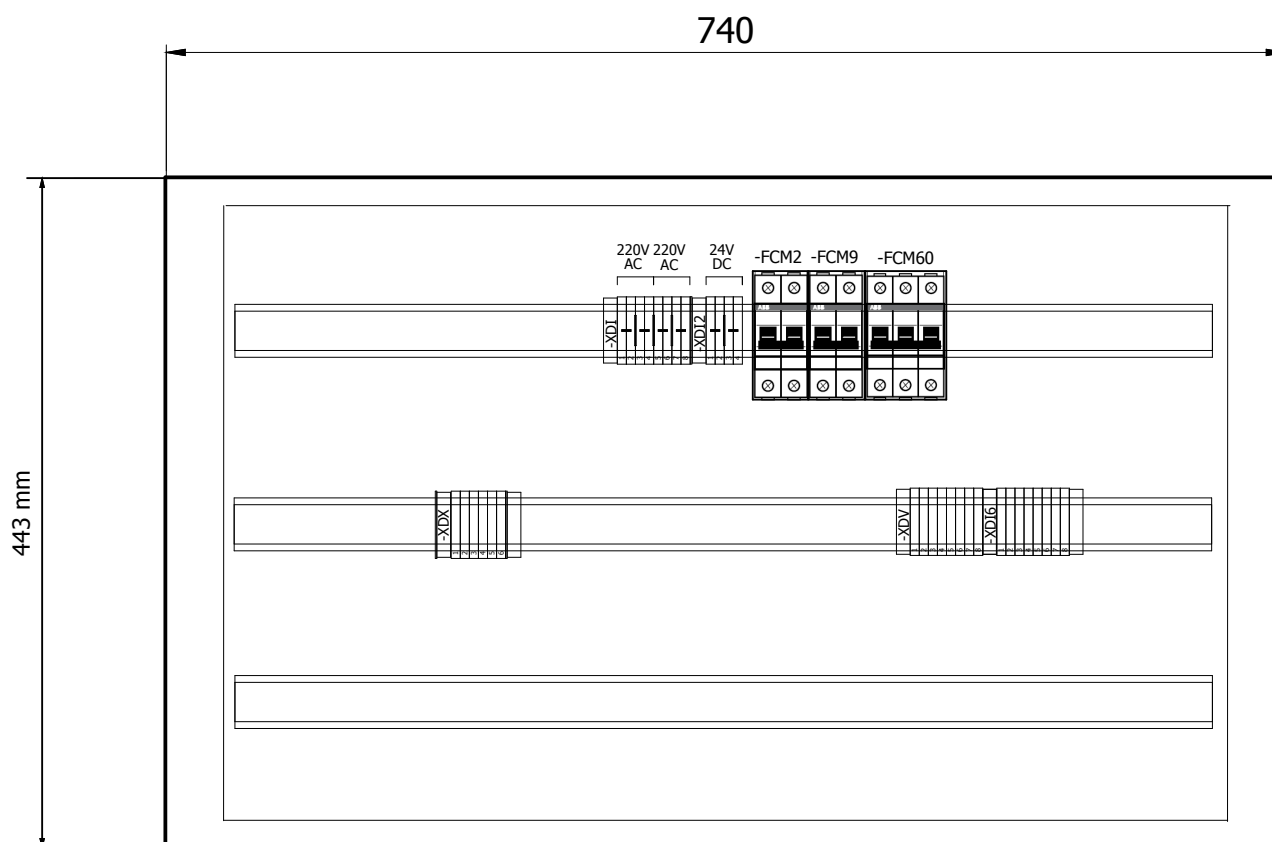


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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	LV BOX LAYOUT MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H03
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	17.01.19	YR	Checked	AW				Drawing Nbr.	18.042-MNA-308.12	Sh. 10
Ind.	Revision	Date	Name	Approved	AI	Customer Doc. Nbr.			Cont. 16		
Based on: BE318042						PT. ABB Sakti Industri					

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	1	2	3	4	5	6	7	8								
A	Summarized parts list							IDABB_F02_005-IDABB								
								Carry over: 0.00								
B	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS								
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.7:E								
	-FCM2	MCB S202-C2 MCB. 2P, 2A			ABB	2CDS252001R0024	1	/7.2:C								
	-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	1	/7.6:C								
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/8.4:C								
	-PGV	Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM			RISHABH	DPM PGD3V	1	/7.2:D								
C	-XTV1	Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSPTT-04C	1	/8.5:C								
D																
E																
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST	Scale		= H03		
					Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 308		
	00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri	MVD-32 20kV Tank Farm WICA	SCHEMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-308.12	Sh. 11		
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042				DRC BUS PT	Customer Doc. Nbr.		Cont. 16	
	1		2		3		4		5		6		7		8	

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[illegible]

F		E		D		C		B		A	
REMARK		CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK	
==MVD-32=H03+-XDI6			-PREVIOUS_PNL-XDI6:2		1 ●	PT 2.5		-PGV:1			
					2 ●	PT 2.5		-NEXT_PNL-XDI6:1			
					3 ●	PT 2.5			-PGV:3		
					4 ●	PT 2.5			-NEXT_PNL-XDI6:2		
					5 ●	PT 2.5			-PGV:5		
					6 ●	PT 2.5			-NEXT_PNL-XDI6:3		
					7 ●	PT 2.5			-PGV:6		
					8 ●	PT 2.5			-NEXT_PNL-XDI6:4		
REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.											
Ind.		Revision		Date	Name	Approved	Based on:		BE318042		
							PROJECT TITLE		MV Switchgear for MNA Serang Plant		
							ABB EPMV		Terminal diagram ==MVD-32=H03+-XDI6		
							PT. ABB Sakti Industri		Scale		
									Customer		
									Drawing Nbr.		
									Customer Doc. Nbr.		
									= H03		
									& 308		
									Sh. 14		
									Cont. 16		

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